

Q1. Dataset : Hospital patient Segmentation

patient ID	Age	monthly medical expense	Health Risk score
P ₁	28	12000	65
P ₂	55	25000	40
P ₃	42	20000	75
P ₄	30	11000	70
P ₅	60	28000	35

SSE values

k SSE

1 600

2 350

3 220

4 200

5 190

Q. A health care organization use k-Means clustering to group patients based on their medical expenses and health risk scores. Interpret the elbow point from the SSE values provided and determine the optimal no. of clusters. Discuss how selecting this no. of clusters affect patient segmentation and health care decision-making.

K	SSE	Reduction from previous k
1	600	—
2	350	250
3	220	130
4	200	20
5	190	10

when $k=1, 2, 3$ the SSE value is reduced much

$k=4, 5$ the decrease by 130.

This variation happens when value is 3

elbow method suggest value of $k=3$

∴ Based on health risk score 3 clusters are form

2. Moderate - expense by moderate risk
3. High risk with high expense

Impact on health-care decision making

- low risk - prevent care and routine monitoring
- Moderate - Regular follow up & checkups
- High risk - Intensive monitoring

2. A retailer applies Hierarchical Agglomerative Clustering to customers orders using appropriately encoded product - category information and standardized order value. A dendrogram is generated. Interpret how clusters are progressively formed and determine the appropriate no. of clusters by selecting a suitable cut-off level on the dendrogram. Explain how the selected clusters can help understand customers

Sol:-)

1. Calculate the distance b/w clusters
2. Merge the two closest clusters.
- 3.

order ID	product category	order value
O_1	electronics	55000
O_2	clothing	25000
O_3	electronics	57000
O_4	furniture	30000
O_5	clothing	27000

$O_1 + O_3$ forms cluster 1 - electronics

$O_2 + O_5$ forms cluster 2 - clothing

O_4 forms cluster 3 - furniture

U5

130

20

1800

A mobile application company uses the HAC Company to segment users based on their activity patterns. Interpret how users are assigned to clusters using probability-based membership values. Discuss the advantages of soft clustering over hard clustering for customer behaviour analysis.

Step 1:- Identify the features

We have 3 features for each user:

- Usage Time (minutes)
- Login frequency
- purchase Amount (₹)

Step 2:- observe the data

look at the user's behaviour

U_1, U_3, U_5 :

- High usage
- High login frequency
- lower purchase amount

so, they form one similar group

U_2, U_4 :

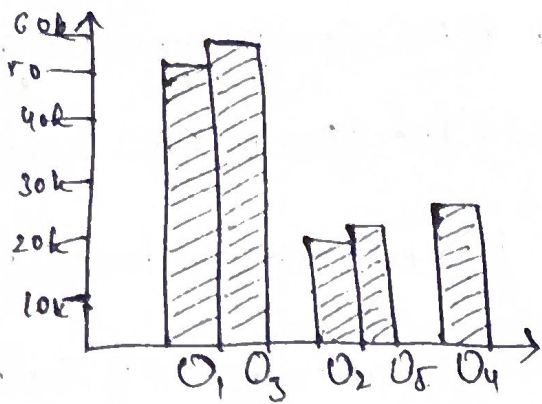
- low usage
- low login frequency
- High purchase amount

so, they form another similar group

Step 3:- Compare the two groups.

This variation happens ~

Dendrogram:-

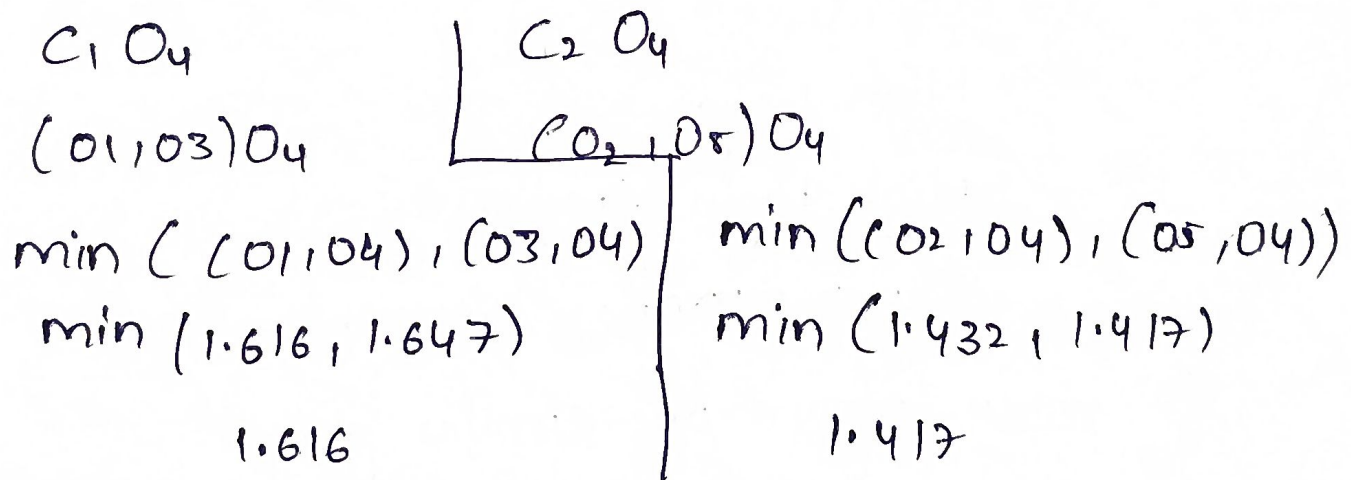


But the optimal no. of clusters cannot be determine by the given data.

Step ÷ 1

1.) one-hot encoding

Order	electronics	clothing	furniture
01	1	0	0
02	0	1	0
03	1	0	0
04	0	0	1
05	0	1	0



Q3: Dataset : Mobile App User Behaviour

user	Daily usage Time	log in frequ	in-App purchase
U ₁	120	18	2000
U ₂	60	5	5000
U ₃	140	22	1500
U ₄	80	4	6000

Features

G1 : U1, U3, U5

G2 : U2, U4

usage

High

low

logins

High

low

purchase

low

High

There is a clear difference between the two groups.

step 4: Determine No. of clusters.

Since these are two distinct behavioral patterns

we choose:

$$k=2$$

Therefore:

No. of clusters = 2

step 5: Name the clusters.

cluster 1 - Highly engaged, lower-purchase users

→ U1, U3, U5

cluster 2 - less engaged, high-purchase users.

→ U2, U4

step 6: Final Result

$$k=2$$

cluster 1 = {U1, U3, U5}

cluster 2 = {U2, U4}

I analyzed three features - usage, log in, frequency and purchase amount. The user frequency shows two clearly different behavioural patterns.

Therefore the appropriate number of clusters

Q4 Dataset: Agricultural crop monitoring

Farm	Rainfall (mm)	soil fertility score	crop yield (tons)
F ₁	300	8	85
F ₂	500	4	70
F ₃	400	6	80
F ₄	280	9	90
F ₅	520	3	65

An agricultural research centre performs clustering on farm data before and after applying normalization. Interpret the difference observed in the clustering results. Explain how normalization influence cluster formation and improves the reliability of agricultural.

Step 1: Calculate distances before normalise

We use Euclidean distance

$$d(P, Q) = \sqrt{\sum (P_i - Q_i)^2}$$

for example,

$$d(F_1, F_4) = \sqrt{(300 - 280)^2 + (8 - 9)^2 + (85 - 90)^2}$$

$$= \sqrt{426}$$

$$\approx 20.64$$

Similarly the important distances are

pair	Distance
F ₁ - F ₄	20.64
F ₂ - F ₅	20.64
F ₁ - F ₃	100.14
F ₂ - F ₃	100.12
F ₃ - F ₄	120.45
F ₃ - F ₅	120.97
F ₄ - F ₅	241.82

HAC before normalization
using single linkage the smallest distance is:

$$F_1 - F_4 = 20.64$$

Next, F_2

and

$$F_2 - F_5 = 20.64$$

Therefore,

$$C_1 = \{F_1, F_4\}$$

$$C_2 = \{F_2, F_5\}$$

$$C_3 = \{F_3\}$$

The smallest distance are $F_1 - F_4$ and $F_2 - F_5$

Step 4: HAC clustering

At 3 clusters

$$C_1 = \{F_1, F_4\}$$

$$C_2 = \{F_2, F_5\}$$

$$C_3 = \{F_3\}$$

$\therefore$ The similar form are grouped together

Step 5: Interpretation

Before normalization rainfall dominates it because its numerical range is much larger. After normalization, all features are scaled to 0-1, giving rainfall, soil fertility and crop yield balanced influence.

Q5. Data: employee performance evaluation.

Employee	productivity	Score Attendance (%)	Skill Assessment Score
E1	88	92	84
E2	62	78	68
E3	78	88	72
E4	55	65	58
E5	93	96	90

with low expense

Silhouette scores

model	score
$k=2$	0.48
$k=3$	0.68
$k=4$	0.52

A company clusters employees based on productivity attendance and skill assessment scores. Analyse the silhouette scores obtained for different clustering models and determine the most suitable no. of clusters. Justify your answers and explain how the selected model contributes to effective employee performance evaluation.

Given Silhouette Scores

Model	Score
$k=2$	0.48
$k=3$	0.68
$k=4$	0.52

Analysis:-

The silhouette score measures the quality of clustering. A higher score indicates better-defined and well-separated clusters.

- $k=2$; Score = 0.48 $\rightarrow$ moderate
- $k=3$; Score = 0.68 $\rightarrow$ Best clustering
- $k=4$; Score = 0.52 $\rightarrow$ moderate clustering

Since $k=3$ has the highest silhouette score (0.68), it is the most suitable model.

Justification :-

$k=3$ provides better similarity among employees within the same cluster and better separation between different clusters. It can effectively group employees based on their productivity, attendance and skill assessment.

Conclusion :-

Best no. of clusters = 3

The selected $k=3$ model can help the company identify different employee performance groups, such as high, medium, and low performers. This supports better performance evaluation, training, recognition and employee development decisions.